



CAPSTONE PROJECT GUIDE (“CPG”)

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Testimonials from previous Bridge students

"My study group made Tuck Bridge one of the most incredible experiences of my life. I loved working hard and presenting our final capstone project together."

"The capstone project taught me how to perform business research and create a professional slide deck."

"Working with my small group on the capstone project was my favorite part of the Bridge program."

Chapter 1. Overview & Objectives

I. What is the Tuck Business Bridge Capstone Project?

A “Capstone Project” is a multifaceted assignment that serves as a culminating intellectual experience for students, typically at the end of an academic program. Capstone projects vary from program to program but are often required to provide students with the opportunity to use what they have learned and apply it to a specific area of professional practice.

The Capstone Project allows students to apply the skills they acquire in the classroom to analyze and value real-world business. Each team will select a publicly traded company, analyze its business, and determine how much the company is worth. At the end of the Bridge program teams will present their recommendations and conclusions to professionals from investment banks and consulting firms.

II. What is the primary objective of the Capstone Project?

Students will choose a public company and derive an “intrinsic value” for the company. A commonly accepted definition of intrinsic value is “The price a rational investor is willing to pay for an investment, given its level of risk.” When you use intrinsic value you are following a key tenant of Berkshire Hathaway CEO Warren Buffett’s philosophy: “Never invest in a business you cannot understand.”

Intrinsic value measures the value of an investment based on its cash flows. Where market value tells you the price other people are willing to pay for a company, intrinsic value generates a company value based on an objective analysis of a company’s current and expected future performance and risk. Bridge students will learn and apply the discounted cash flow (DCF) method for calculating the intrinsic value of

the public company they chose. To determine value using DCF students need to estimate future cash flows and select an appropriate discount rate. Through the DCF, they will learn both fundamental drivers of company value as well as the nuanced competitive and economic factors that influence those fundamental drivers.

This project is not meant to be a stock picking exercise (even though many “stock pickers” and Wall Street analysts would conduct a similar investigation). Rather, it is the kind of analysis that would be conducted by a buyer interested in acquiring the entire business.

Based on their extensive company analysis, each team will make a recommendation to such a potential acquirer. The recommendation will include the purchase price that the investor would be willing to pay for all the company’s equity. Students should also strive to offer suggestions as to how the new owner might improve the company’s performance going forward.

III. Why is a company valuation using a DCF ideal for the Capstone Project?

A company valuation using a DCF focuses students’ attention on the financial, economic, and competitive aspects of an existing, publicly traded company in order to undertake a valuation of its equity. It is designed so that students have an opportunity to apply most aspects of the Bridge curriculum. Financial accounting, corporate finance, marketing, economics, strategy, spreadsheet

modelling, business communications, and organizational behavior will all be useful – critical in fact – as they conduct their analyses and present their recommendations.

A company valuation using a DCF blends the application of timeless, fundamental concepts from various academic areas with the need to use critical thinking and analytic skills. For example, teams must arrive at an assumption about their company's future sales growth.

This is a fundamental step in forecasting free cash flows, an important component of a valuation. However, the ever-changing economic and strategic environment in which companies operate means that choosing a forecast assumption requires a thorough and pragmatic analysis of the current business environment. In short, the project is both timeless and timely.

IV. What support will students receive in completing the Capstone Project?

Students will have many resources available to help them complete the project. Aside from the tools that they will acquire in the classroom, teams will work closely

with a designated Bridge Associate (Tuck MBA student) who will guide them throughout the process.

Students will also have access to Tuck's library and its experienced research librarians who will support them as they collect data and conduct research. Select examples of project presentations and slides will also be made available to current Bridge students. See the Capstone Project 'Resources' tab in Canvas.

In addition, most Bridge faculty will offer one session during the program to assist students in applying the material covered in their course to the project. In addition, each group will have several opportunities to meet one-on-one with a Bridge faculty member to assess progress and ask questions. Of course, all members of the Bridge faculty will be available to help you answer questions that require specific domain expertise at any point during the program.

V. What will students gain from completing the Capstone Project?

Bridge students will have approximately two weeks to select a public company, gather data, conduct a series of analyses, create a slideshow that clearly and succinctly communicates their findings, and present their recommendations to an authentic audience. Time management, teamwork, goal setting, and communication will be among the essential life skills that this project will both challenge and nurture.

Participation in the Capstone Project will help prepare students for a wide range of career paths. Whether future entrepreneurs, general business managers, investment professionals, consultants, or non-profit board members, this experience will help students become better decision makers and communicators in any business setting.

Chapter 2. Project Milestones

The Capstone Project is designed to be a culminating program experience with a formal presentation delivered to an authentic audience on the final day of the program. However, as the underlying analysis is comprehensive and must be completed within a short period of time, students are encouraged to think of the assignment as a series of milestones that are completed along the way. This will ensure your readiness for the final presentation and provide opportunities to receive guidance from faculty and fellows as your team progresses throughout each milestone.

The milestones briefly described below are designed to help you efficiently complete the project by synchronizing it with the in-class curriculum. This synchronization ensures that concepts delivered in class are applied directly to the project when you are ready to do so. The process of completing a company valuation is not linear.

Expect to make well-informed refinements to your DCF valuation model and conclusions throughout the program and up to the night before the final presentation.

Note that some milestones involve deliverables while others can be viewed as internal team milestones. Refer to Canvas for your program-specific timing of deliverables and a detailed description of each milestone.

I. Project Launch and Select Public Company

Each team will select a publicly traded company to analyze. This will take place very early in the program immediately following the Project Launch. See Chapter 3 of the CPG: Selecting a Public Company.

II. Company Profile

You can begin working on this company profile as soon as you know your approved capstone project company. Your team will start learning about your company and the environment in which it operates. Tuck's librarians will be available to help you with data sources and research techniques that you can use for this assignment (and throughout the project). Consider the following narrative for your profile:

1. Briefly describe your company's products and customers. If the company operates in multiple major segments, describe them separately.
2. Briefly describe the industry in which the company operates.
 - Who are the company's main competitors? Are they large or small relative to the company? Are there any other major players that the company interacts with (e.g., major suppliers, regulators, producers of complementary products)?
 - What are the challenges and opportunities facing the industry today? Are there significant industry trends that might affect the company's business in the future?

Deliverable: Submit your company profile to Canvas. The profile need not be more than 2 pages and can consist of bullet points.

III. Financial Analysis

Using what you learned in Financial Accounting you are equipped to complete a Financial Analysis of your capstone company. This milestone involves becoming familiar with your company's financial statement data and learning about its revenue growth, costs, and investments.

As part of this analysis, you will download three years of historical income statements, balance sheets and cash flow statements for your company. From there, you will simplify the statements and modify the project template to include your company's historical data from which you will compute some key financial ratios. It is not critical at this point that you understand the trends in your company's historical data, but you should view it as an important starting point for developing future assumptions.

Students will work on this with the Financial Accounting Professor as part of the Financial Accounting course and the Capstone Project Accounting Lab. Discuss insights from this exercise with your Bridge Fellow.

IV. Build 3-Statement Model

Using what you learned in Financial Accounting you are equipped to complete the 3-Statement model for your capstone company. This milestone involves setting some initial assumptions about future revenue growth, costs, and investments. Continue using the same Excel template where you performed your Financial Analysis. From there, you will use those assumptions to build a set of forecasted financial statements.

You will have the opportunity to refine your model as you learn more in your Economics, Strategy, and Marketing classes. These classes will give you the frameworks for thinking through the key assumptions in your model. For this milestone, do not worry about getting these assumptions “right” but start noticing how they might affect your company's ability to generate free cash flow.

Students will work on this with the Financial Accounting Professor as part of the Financial Accounting course and the Capstone Project Accounting Lab.

Discuss insights from this exercise with your Bridge Fellow.

V. Revenue Model Analysis

This milestone involves analyzing the company's revenue and, when possible, operating income.

Most companies break down their revenue (and, often, their operating income) in “reportable (or operating) segments”. Companies have a lot of flexibility on how they choose their reportable segments. Sometimes they use obvious ways – e.g., retail and wholesale, in-store and online, domestic and international etc. – and sometimes they use segments that match how they manage their business.

For example, Apple breaks down revenue in two ways: by geography (Americas, Europe, Greater China, Japan, and of Asia Pacific) and by product and service (iPhone, Mac, iPad, Wearables, Home and Accessories, and Services). Amazon, on the other hand, uses a simple breakdown: North America, International, and AWS.

The objective here is to disaggregate revenue to “building blocks” so that you develop deeper insights about the company’s operations and be able to project future revenue not as a one-line item, but on a segment-by-segment basis (the idea being that not all segments have the same growth and profitability profiles). In other words, to not treat revenue as a “black box” but actually develop insights on what “drives” it.

If the company does not analyze revenue into reportable segments, it’s an opportunity for you to come up with your own way of projecting future revenue. For example, for a retailer, total revenue could be the product of Number of Stores x Average Revenue Per Store. (Remember, $\text{Revenue} = \text{Price} \times \text{Quantity}$). Then, you can make separate assumptions about how the number of stores and the average revenue will fare during the projection period.

As you analyze the revenue and operating breakdown of the past three years, think about the following questions:

- Where does the company capture the greatest margins? Are these sustainable?
- Can you identify what “drives” revenue in each reportable segment? For example, it could be the number of users, number of stores, presence in growing markets etc.
- How would you evaluate the company’s pricing power?

Deliverable: Analyze the revenue and the operating (where available) by reporting segment, for the historical 3-year period. Submit a one-pager with your analysis by year, along with your responses to the questions above.

VI. Economic Analysis

This deliverable will help prepare teams for both the capstone lab and your final company assessment. Please, upload a written document that contains a series of bullet points addressing the questions listed. These bullet points should support the narrative underlying key financial forecast assumptions in your model.

Please note you will complete this assignment for each course (Economics, Strategy and Marketing), and each course’s questions should be addressed separately.

- Briefly discuss the competitive environment in your industry. What are the main products? Who are the target customers? Who are the main competitors? What is your company’s market share? Are there barriers to entry? Is there a scarce resource and does your company control it? How are prices determined – is your company a price taker or do they have the ability to control prices?
- What drives revenues in your company’s industry? What are the drivers of demand? Is demand elastic or inelastic? Recall total revenue is $TR = P \times Q$. Is your company well positioned to take advantage of changes of demand, and/or does your company have the ability to charge above-competitive prices? What strategies does your company employ to maintain or gain market share?

- Costs are just as important to profit as revenue, and costs can change and be predicted just like revenues. What are the main costs for your company and how do they vary based on growth? Depending on the industry, you might consider labor market costs, cost of production inputs (raw materials, minerals, etc.) and other investments (factories, tankers, etc.). Combine your discussion of costs with your discussion of revenue and make a prediction about profit growth.

Students will receive more specific guidance and feedback from the Economics Professor as part of the Capstone Project Joint Lab. Discuss insights from this exercise with your Bridge Fellow.

VII. Strategic Analysis

This deliverable will help prepare teams for both the capstone lab and your final company assessment. Please, upload a written document that contains a series of bullet points addressing the questions listed. These bullet points should support the narrative underlying key financial forecast assumptions in your model.

Please note you will complete this assignment for each course (Economics, Strategy and Marketing), and each course's questions should be addressed separately.

- Identify your company's key resources and capabilities.
- Categorize your company's key resources using the tangible/intangible/human typology and categorize your company's key capabilities using the functional area classification.
- For each key resource and capability separately, analyze whether it meets the VRIN criteria. First ask whether each resource or capability meets the VR criteria for competitive advantage. If the resource or capability meets the VR criteria, then ask whether the same resource or capability meets the IN criteria for sustainable advantage.
- Also analyze whether your company has any cross-functional capabilities or bundles of intertwined resources and capabilities that meet the VR criteria and, if so, whether they meet the IN criteria.
- Considering all of your company's key resources and capabilities together, how sustainable a competitive advantage does your company derive from its resources and capabilities, ranging from no current competitive advantage to a strongly sustainable competitive advantage?
- What implications does your answer have for the narrative underlying the key financial forecast assumptions in your model?

Students will receive more specific guidance and feedback from the Strategy Professor as part of the Capstone Project Joint Lab. Discuss insights from this exercise with your Bridge Fellow.

VIII. Marketing Analysis

This deliverable will help prepare teams for both the capstone lab and your final company assessment. Please, upload a written document that contains a series of bullet points addressing the questions listed. These bullet points should support the narrative underlying key financial forecast assumptions in your model.

Please note you will complete this assignment for each course (Economics, Strategy and Marketing), and each course's questions should be addressed separately.

The Marketing Mix:

- How is your company addressing the changing wants and needs of their target market with their products/services, communication, pricing, etc.?

Brand Equity:

- Is your company finding ways to improve their brand equity? (Associations, Loyalty, Perceived Value, Awareness). How does your Net Promoter score reflect this?

Marketing Environment:

- Immediate Environment: How does your company address two of the Immediate Environmental factors.
- Macro Environment: How is your company responding to three of the macro environmental factors as they relate to market movement?

Target Market:

- Is your company's target market sustainable for the future? How are they addressing the shifting market realities?

Students will receive more specific guidance and feedback from the Marketing Professor as part of the Capstone Project Joint Lab. Discuss insights from this exercise with your Bridge Fellow.

IX. Valuation Narrative

Building on the Economics, Strategy, and Marketing bullet point analyses you completed earlier, develop a coherent narrative to support the key assumptions in your forecast that you began to work on while building your 3-statement model. Each team will present during the Capstone Project Joint Lab regarding what they learned in Economics, Strategy, and Marketing and how they are applying the concepts to their overall company analysis. Each team will receive feedback about the ideas in their presentation from both their peers and faculty.

This is a means of encouraging you to think about the narrative surrounding your assumptions early and identify gaps in your research. See Chapter 5 of the CPG: Reflecting Your Company Analyses in Your DCF Model.

X. Build DCF Model

This milestone involves completing the DCF model tab in the capstone Excel template.

To complete this tab, you will need to calculate a cost of capital for your company using what you learned in Corporate Finance, determine a terminal growth rate, and input the number of shares outstanding as well as the current level of cash and debt.

You will need the cost of capital to discount the free cash flows that you calculated in your 3-Statement model. As your valuation involves many assumptions that are uncertain, you will complete this initial DCF analysis which will then involve a series of refinements to your initial assumptions. Ultimately, you

will decide on a final set of assumptions. Think about which sensitivity analyses you will include in your presentation.

Students will work on this with the Finance Professor as part of the Capstone Project Finance Lab. Discuss insights from this exercise with your Bridge Fellow. See Chapter 4 of the CPG: Building a DCF Model for Performing a Company Valuation.

XI. Executive Summary Lab & Prepare Presentation Slides

Your team will consider all of the different analyses it has conducted and consolidate its work into an Executive Summary. Use the frameworks from your Business Communication class. You will practice delivering your Executive Summary to the Business Communications Professor during the Business Communication Executive Summary Labs. Work closely with your Bridge Fellow as you create your narrative. See Chapter 7 of the CPG: 'Preparing a Persuasive Presentation'.

XII. Rehearse Capstone Presentation

Your team will rehearse its presentation with selected members of the faculty. You will be encouraged to hone your presentation skills and polish your PowerPoint slides. You should work closely with your Bridge Fellow to ensure that your message is clear, that your recommendations are well supported, and that you have considered important sensitivity analyses.

XIII. Deliver Final Presentation to Panel

Your team will formally present its conclusions to a panel of professionals during the final morning of the program. You will have up to 20 minutes to present your analysis, leaving approximately 10 minutes for Q&A with evaluators immediately thereafter.

Chapter 3. Selecting a Public Company

Access the Google doc posted in Canvas to select a public company for your group to analyze. Be prepared to submit your choices by the end of day after the Project Launch.

You must submit up to three choices listed in order of your preference. One of your company choices will be approved by a faculty member and noted in the Google doc. See Chapter 3 of the CPG: Selecting a Public Company.

Be thoughtful when choosing your company as you will spend a lot of time learning about its business. No more than one group will be allowed to analyze a company. From this point forward, you should place your chosen company in the context of each new skill you acquire in the classroom during the Bridge program.

Guidance for Selecting a Public Company

In Canvas, we provide an Excel file that contains a list with publicly traded companies in the U.S. This Excel file also contains some limited information about each company's relative valuation and industry.

You are not required to choose a company from this list (and there are more publicly-traded companies not included in the list), but it may be useful in generating some discussion among your group about which company to choose. Keep in mind that you will spend a lot of time thinking about your company so settle on something that you are interested in enough to conduct substantial research.

Column	Variable	Variable Significance
A	Company name	
C	P/E ratio (as of May 2026)	Indicator of relative valuation
D	Economic sector	
E	Industry name	

The ideal company to select for your capstone project would be a 'pure play' company. Pure play companies focus their efforts and resources on only one line of business, i.e., they specialize in a specific niche or have little to no vertical integration. As such, their stock performance correlates highly to the performance of its industry or sector. This focus makes it easier to analyze your company and to determine its beta, which in turn determines its weighted-average cost of capital (see Corporate Finance material).

However, pure play companies are often difficult to identify among public companies, which are generally large and diversified businesses. When they do exist, they are not often the types of well-known companies that we are familiar with, and they may not have close competitors. For example, Stitch Fix would be considered a pure-play online styling service. But the company does not have any true peers, or at least not any that are publicly traded. The closest analog may be Trunk Club, a similar service owned by Nordstrom, but we cannot compare the two on valuation because Trunk Club is part of a larger, diversified company. As a result, identifying a pure play company may sometimes require a broader definition of niche such as "consumer-facing online retailers". This might

lead you to choose a company like Wayfair (home furnishings e-commerce) or Etsy (marketplace for artisan-made products).

The bottom line is that the more focused your company is, whether it be product, customer, or geographic focus, the easier it will be to analyze in the short period of time that you must complete the project. The following are some things to keep in mind that will make your analysis more difficult:

- Less than 3 years of history as a public company, such as a recent initial public offering (limited historical data to assess trends); all of the companies on the list that we provide have 3 years of history as a public company
- Recently undergone a significant merger, acquisition or divestiture (difficult to analyze historical data to assess trends because business has been reorganized)
- Operates in many business sectors (more industries/products to understand)
- Operates in many geographic markets (more markets to understand)
- A limited number of (or no) true peer companies for comparison

Your choice will be a trade-off among many things. You may want to challenge yourself with a large, diversified company whose name you are familiar with. Or you may want to choose a smaller company operating in an industry that you know very little about, such as simulation software. And remember, some industries that may not seem that exciting have recently become very interesting considering the technological and geopolitical context. Good luck!

Chapter 4. Building a DCF Model to Perform Company Valuation

We provide an example to use as a starting point for building a Discounted Cash Flow (DCF) model. To emphasize that a DCF model derives from a three-statement model, the example includes two tabs: '3 Statement Model' and 'DCF Model'. The example proposes a useful structure for a DCF model and, through its design, illustrates the appropriate flow of information. One size does not fit all when it comes to building a 3-Statement or DCF model because every company is unique. Expect to make modifications out of necessity (because your company has something unique about its financial statements) or because your group prefers to analyze a certain aspect of your company's financial data in more or less detail than the example proposes.

What is a 3-Statement Model?

A 3-statement model is the most basic financial model, and the model on which most other advanced models, including a DCF model, are built. For example, executives inside a company will use a 3-statement model to make decisions about raising capital (debt and/or equity), making acquisitions (businesses and/or assets), growing the business organically (e.g., opening new stores, entering new markets, etc.), selling or divesting business units, budgeting and planning for the years ahead, capital allocation (priority of which projects to invest in), valuing a business, or financial ratio analysis. Thus, learning how to build a 3-statement model provides a solid foundation for using financial data in a variety of ways in the future.

Figure 1: The 3-Statement Model

3 Statement Model USD thousands, except per share data	Historical -->			Forecast -->				
	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Balance Sheet Check	OK	OK	OK	OK	OK	OK	OK	OK
Assumptions & Value Drivers								
Income Statement								
Balance Sheet								
Cash Flow Statement*								
Supporting Schedule								

As the name implies, in a 3-statement model the three statements (income statement, balance sheet, and cash flow statement) are all dynamically linked¹. The objective is to set it up so that the accounts are connected, and a set of assumptions drive changes in the entire model. Setting up the 3-statement model requires a solid foundation in accounting and spreadsheet modelling. Determining the set of assumptions themselves requires a solid foundation in finance, economics, strategy, and marketing as they rely on your understanding of the company and its value drivers.

See Chapter 6, Reflecting Your Analyses On the DCF Model for further discussion of identifying key value drivers and linking them directly to your assumptions.

¹ Any supporting schedules or charts & graphs are optional. Use supporting schedules to simplify, or make transparent, some of the calculations in the financial statements. Use charts & graphs to better visualize, interpret, or communicate data during the historical or forecast periods

In the example shown above in Figure 1, financial data from 2023A to 2025A denote the historical period, with the 'A' denoting actual data. Financial data from 2026E to 2030E denote the forecast period, with the 'E' denoting expected data. The process of building a 3-statement model can be summarized in four steps. You will practice building a 3-statement model in the Financial Accounting course.

Step 1: Enter actual financial data for your company during the historical period

Step 2: Examine a set of key metrics during the historical period ("financial analysis")

Step 3: Set values for assumptions during the forecast period

Step 4: Use assumptions to generate expected financial data during the forecast period

How do I build a DCF model using a 3-Statement Model?

The first thing to spotlight about the DCF model that you will build during the Bridge program is that it is a model built for a specific purpose: To value a company's equity based on the present value of the company's expected future cash flows. Given this purpose, certain information from the 3-statement model will be emphasized and play a key role in the DCF model; the most significant of which will be the information required for the 'Calculation of Free Cash Flows'². See next chapter, Chapter 5.

Figure 2: The DCF Model

Forecast -->					
Discounted Cash Flow Model					
USD thousands, except per share data					
	2026E	2027E	2028E	2029E	2030E
Calculation of Free Cash Flows					
Calculation of Weighted-Average Cost of Capital (WACC)					
Calculation of Terminal Value (Perpetual Growth Method)					
Calculation of Equity Value and Price Per Share					
Sensitivity Analysis					

² Other information such as debt and cash on the balance sheet will be used as well. See Chapter 5.

Chapter 5. Determining Equity Value Using the DCF Model

The DCF model example is divided into four main sections. Each section is described below. The exit multiple method, which is often used in practice to complement a valuation using the DCF approach, is optional.

1. Calculation of Free Cash Flows

The calculation of (unlevered) free cash flows (FCFs) to both debt and equity holders in the forecast period is the portion of the DCF model that links directly back to the 3-statement model. The inputs used to calculate these FCFs are summarized below for each forecast period. Then, each FCF from 2026E through 2030E is discounted back to the end of 2025A (the end of the historical period) and summed (See 'Calculation of Weighted-Average Cost of Capital' in the next section of the DCF model). A typical forecast period is five years but could be longer depending on the company.

Discounting is the process of determining the present value of a stream of cash flows that are to be received in the future. A dollar is worth more today than it would be worth tomorrow and you will learn in Corporate Finance how to think about the relation between a dollar received today and a dollar received in the future.

Forecast -->					
Discounted Cash Flow Model					
USD thousands, except per share data	2026E	2027E	2028E	2029E	2030E
Calculation of Free Cash Flows					
	1	2	3	4	5
Operating Income (also called Earnings Before Interest and Tax, or EBIT)	\$691	\$767	\$820	\$844	\$834
Operating Income * (1-Tax rate)	\$518	\$575	\$615	\$633	\$625
PLUS: Depreciation expense	\$202	\$235	\$263	\$284	\$295
MINUS: Capital expenditures	(\$238)	(\$276)	(\$309)	(\$334)	(\$347)
MINUS: Change in Net Working Capital	(\$165)	(\$68)	(\$64)	(\$54)	(\$38)
(Unlevered) Free Cash Flows (FCF)	\$317	\$465	\$504	\$529	\$536
Present value of FCFs	\$288	\$382	\$376	\$357	\$328
Sum of Present value of FCFs	\$1,730	<<<--- See 'Calculation of WACC'			
Check	\$1,730	<<<--- Using NPV formula in Excel to check			

All the assumptions needed to estimate FCFs are included in the Assumptions & Value Drivers section of the 3-statement model. Forecasting Operating Income requires assumptions about revenue growth and costs or profit margins. Operating Income*(1-tax rate) requires an assumption about the tax rate the company will be subject to on its future Operating Income. Depreciation Expense and Capital Expenditures require assumptions about future investments in tangible assets.

Finally, the change in Net Working Capital requires assumptions about investments in inventory, receivables, or payables the company will need to maintain going forward to run and to grow or expand its business. For example, if holding more inventory is required to support the anticipated level of revenue growth, then this additional investment will use cash.

These assumptions will come from analyzing your company's historical financial data and thinking about its likely path forward. The mechanics for inputting and analyzing your company's historical financial statement data and forecasting free cash flows are covered in the Financial Accounting course.

2. Calculation of Weighted-Average Cost of Capital (WACC)

To determine the present value of your company's future free cash flows, you will need a discount rate. A discount rate, or the weighted-average cost of capital (WACC), is company-specific. The example calculation shows you the structure of the calculation and how the WACC is used in the DCF model to discount both the free cash flows during the forecast period and the terminal value (See 'Calculation of Terminal Value' in the next section of the DCF model). The theoretical underpinning for calculating and discounting future cash flows, determining the WACC, and appropriate inputs to calculate the WACC are covered in Corporate Finance.

Calculation of Weighted-Average Cost of Capital (WACC)					
rE	11.4%	rf + B*MRP (CAPM)			
E (marketcap)	\$15,330	# of shares outstanding x market price per share			
E+D	\$17,330				
rD	2.8%	The company's interest rate on existing debt or the yield of corporate bonds with similar credit rating (e.g. AAA/Aaa)			
1-T	75%	1 - tax rate			
rf	2.3%	Yield on 30-year treasury bond			
B	1.30	beta			
MRP	7.0%	Market risk premium			
WACC	10.3%	$rE*[E/(E+D)] + rD*(1-T)*[D/(E+D)]$			

3. Calculation of Terminal Value (Perpetual Growth Method)

When building a DCF model there are two major components: (a) the present value of cash flows generated during the explicit forecast period (section 1 above) and (b) the present value of the terminal value (calculated here). The terminal value calculation estimates the free cash flows generated, and hence the value of a business, after the explicit forecast period ends. The terminal value is a critical part of the DCF model, as it typically makes up a large percentage of the total estimated value of a business.

Calculation of Terminal Value (Perpetual Growth Method)					
Terminal Cash Flow					
Free cash flow in 2029E	\$536				
Terminal growth rate (g)	4.0%	<<<--- One possibility is to use the long-term (U.S.) GDP growth (may depend on firm and industry)			
Terminal cash flow	\$557				
Terminal value (TV)	\$8,803	<<<--- Terminal cash flow/(WACC - g)			
Present value of TV	\$5,386				

The perpetual growth method assumes the business will continue to generate free cash flows at a constant rate forever. The terminal cash flow is determined by applying this constant rate to the last cash flow of the explicit forecast period. The terminal value is then calculated by dividing the terminal cash flow by the difference between the WACC and the terminal growth rate (an assumption). Finally,

the terminal value is also discounted back to the end of 2025A (the end of the historical period). The theoretical underpinning for calculating and discounting the terminal value are covered in Corporate Finance.

4. Calculation of Equity Value and Price Per Share

When you value a business using unlevered free cash flows in a DCF model, you are calculating a firm's enterprise value. Simply put, the enterprise value is the entire value of the business, without considering its capital structure, and equity value is the total value of a business that is attributable to the shareholders. For this reason, there will be an additional step required to arrive at equity value from enterprise value.

Calculation of Equity Value and Price Per Share				
Enterprise value	\$7,115	<<<--- PV(FCFs) + PV(TV)		
Plus: Cash	\$1,554			
Minus: Debt	\$2,000			
Equity value	\$6,669			
# of shares outstanding	1,750	<<<--- The number of shares should be expressed in the same units as every other figure, i.e. if the FCF amounts are in thousands, the number of shares should be in thousands too		
Price per share	\$3.81			
Current Market Price	\$8.76	<<<--- Update with the company's current market price		

Enterprise value represents all a company's discounted future cash flows, including those generated during the explicit forecast period as well as the terminal value. Note that in this example, the terminal value is equal to approximately 76 percent of enterprise value (\$5,386/\$7,115), illustrating how sensitive a DCF valuation can be to assumptions about g and WACC.

Once you estimate the enterprise value, you then calculate the equity value by subtracting off the net debt (i.e., debt minus cash) on the balance sheet at the end of the forecast period. Note that the company in this example does not have any debt. Not all companies use debt.

Chapter 6. Reflecting Your Analyses On the DCF Model

The previous chapter illustrates the mechanics of calculating free cash flows (FCFs) during the forecast period, given a set of assumptions about future revenues, costs, and investments, etc. These assumptions are fixed in the ‘Assumptions and Value Drivers’ section of the 3-statement model. Where do these assumptions come from? They come from your company analyses, which include financial, economic, strategic, and marketing analyses. When completing each of these milestones, with guidance and input from Bridge faculty and your Bridge Fellow, be vigilant of tying what you learn directly to your valuation model and using the frameworks you learn in the classroom.

This chapter provides an overview of how to turn your work into a ‘story’ that begins with your value estimate and supports it with insights and analyses. The example company is Stitch Fix (SFIX). Note that this illustration is brief relative to the in-depth analyses that your team will conduct. The objective is to provide a roadmap, with real-world examples, so you can see how these analyses tie directly to company value.

The Roadmap

A useful way to approach the Capstone Project is to think of your work before the final presentation as building to a single number that is the company valuation – i.e., “Our team believes that Stitch Fix’s equity should be valued at \$3.293 billion, which is lower than the current market valuation.” The ‘story’ to your client, however, should begin with this number and support it with the insights that you obtain from conducting company analyses – i.e., “We arrived at our \$3.293 billion value estimate by identifying and incorporating three key drivers of value creation into our valuation model.” Structure your presentation as a value driver narrative. Figure 1 illustrates one possible narrative for Stitch Fix.

Figure 1. Value Driver Narrative for Stitch Fix (SFIX)

Value Drivers → Fundamentals ↓	Example 1: Mass Customization	Example 2: Artificial Intelligence Augmentation	Example 3: Vendor-Managed Inventory
Revenue			
Operating Cost			
Investment			
Risk			
Growth	↓	↓	↓

First, note that the value driver narrative is structured around three value drivers: mass customization, artificial intelligence augmentation, and vendor-managed inventory. A value driver is anything that has the potential to enhance company value, either by increasing cash flows or reducing risk. While there are potentially hundreds of value drivers for any given business (with complex interactions among them) your job is to identify the three most important value drivers for your company and link them to its value creation process in a compelling way.

Second, note that each value driver is anticipated to impact fundamentals: revenues, operating costs, investment, etc. These fundamentals represent the financial data necessary to determine the value of a company. For example, management is expected to take actions in strategic areas that will have the greatest impact on value (the value drivers), which should translate into future revenue growth. These future revenues represent future cash flows that will increase the value of the company.

An Illustration: Developing a Value Driver Narrative for Stitch Fix

Company Profile (brief)

Stitch Fix, Inc. is an online personal styling service that delivers personalized “Fixes” of apparel and accessories to men, women and kids. The company was founded by Katrina Lake and Erin Morrison Flynn in February 2011 and is headquartered in San Francisco, CA. As of December 2021, Stitch Fix’s market capitalization is \$5.805 billion.

The company’s value proposition is to provide busy people with curated boxes of clothing, selected for them by a team of stylists, at an affordable price point. The personal styling service charges a styling fee. Customers fill out their styling profiles online and a Stitch Fix stylist chooses five items to send them. Once received, the customer has three days to choose which items are worth keeping and which ones to return. The “Fix”, as the box is called, is scheduled by the customer to come once a month, every two weeks, every two months, or however often he/she wants it.

Value Driver 1: Mass Customization

Stitch Fix is well-positioned to capture a significant amount of market share within an already high growth market due to their ability to deliver personalization on a massive scale. With a global market value of \$760 billion in 2021 (\$127 billion in the U.S.), apparel, accessories, and footwear (collectively, the fashion segment) are the number one ecommerce sector. Over the next 5 years, online fashion’s expected 7.18% compounded annual growth rate will put the industry at +\$1.0 trillion. Ecommerce apparel growth in 2020 was 29.6% and is the fastest growing fashion segment.

Personalization has become one of the “must-haves” for ecommerce in recent years with 80% of U.S. online shoppers saying they are more likely to make a purchase when retailers offer personalization. Given the fashion industry’s focus on self-expression, it makes sense that modern fashion consumers actively seek personalized experiences. Yet fashion, with its many sizes, collections, and styles, is a lot more complicated and overwhelming to consumers than most other retail categories. While conventional approaches to personalization generally fall short in fashion, Stitch Fix is leading the way in hyper-personalization. Stitch Fix’s strategic focus on driving revenue growth through three key metrics – the number of visitors, the conversion rate, and customer lifetime value – is expected to result in the company delivering between 20% and 25% revenue growth year over year for the next five years, trending towards the industry growth rate of 7% after five years.

In the second quarter of 2021, Stitch Fix delivered \$504 million in revenue, reflecting 12% year-over-year growth, and grew their active client count to nearly 3.9 million, reflecting 12% year-over-year growth. Revenue per client has been relatively stagnant between \$475 and \$500 for the past three years. Stitch Fix is well-positioned to deliver significant growth in visitors, improvements in the conversion rate, and increases in customer lifetime value due to the following factors and innovations:

- With over 3 million monthly visits to their site, Stitch Fix is leaps and bounds ahead of Bespoke Post (their nearest competitor) who see only 0.72 million monthly visits. There is relatively strong awareness of the service.

- Stitch Fix implemented two strategies in the past year to improve conversion within the large prospect population estimated to be in the millions. Fix Preview allows shoppers to view garments in their apparel box before they ship. Direct Buy allows shoppers to purchase a single garment without having to purchase a box. Both programs enable prospects to engage more directly with Stitch Fix and have more agency in selecting the recommended items they wish to receive. This will allow Stitch Fix to appeal to a broader set of clients and drive customer confidence in the company's ability to recommend products.
- Stitch Fix implemented two strategies in the past year to increase customer loyalty, and foster repeat and long-term customers. Shop Your Look allows customers to find items that complement what they've already bought. Similarly, Shop New Colors allows customers to buy previously purchased items in different colors, prints, or patterns.

Value Driver 2: Artificial Intelligence Augmentation

Intelligence augmentation is another conceptualization of artificial intelligence (AI). It focuses on the assistive roles of AI with emphasis on the fact that it is meant to enhance human intelligence rather than replace it. Stitch Fix combines the expertise of approximately 6,000 stylists with the insight and efficiency of artificial intelligence (developed by approximately 100 data scientists) to collect and analyze data on style trends, body measurements, customer feedback and preferences.

This powerful combination arms the human stylists with a culled down version of possible recommendations and supercharges the company's effectiveness and ability to stay competitive. The human input means Stitch Fix can be an arbiter of fashion, rather than just an amplifier because it allows people the chance to interject sometimes surprising or counter intuitive style suggestions that the data might not reveal on its own. This element of surprise is also what hooks customers into the process and makes them more likely to sign up for another Fix.³

Stitch Fix is not the only online retailer that combines technology with a human touch to curate a set of recommended items for the client. Amazon launched its own styling service called Personal Shopper. Even though Amazon is dominant in ecommerce it is playing catch-up to Stitch Fix in personal styling.

Given the large accumulation of data and algorithms within Stitch Fix, the company gets better at predicting what clients will like, as shown by how often clients decide to keep an item they receive in their first Fix. In the first quarter of 2021, 80% of first Fixes to new clients had at least one item that was purchased by the client. This means Stitch Fix has accumulated enough data from its 3.9 million clients that it can accurately predict what new clients will like, when the company has limited data about style preferences. Each client's Fixes improve over time as the company learns more about the customer. This conglomeration of knowledge empowered by data accumulation since 2011 will lower customer acquisition and retention costs, allowing the company to maintain, and even improve its profit margin over time.

³ Stylists have the final say as notes and preferences like "I'm just about to start a new job and need to dress to impress" are so nuanced that algorithms cannot adequately address them. But the stylist is able to truly understand the importance of these contextual comments which is why humans cannot be entirely replaced by AI. People are great at taking the context into account, making subjective judgments and interacting with clients in a personal way that technology cannot.

Value Driver 3: Vendor-Managed Inventory

Stitch Fix, like many retailers, faces risks associated with holding either too little or too much inventory. Inventory can be held in many locations at various stages in the supply chain: at the manufacturer, in distribution centers, in local warehouses, in transit, in retail stores, or in the hands of a customer while he/she considers the merchandise. Effective inventory management is crucial to maintain margins in a competitive retail industry, but it's a delicate balancing act. Too much inventory can lead to discounted merchandise and lower profitability, while too little leads to dissatisfied customers and lost revenue growth.

Stitch Fix has traditionally relied on selling merchandise through a wholesale model, where the company purchases items in bulk and sells them to consumers for a profit. They carry a variety of products that come in many different brands – big and small – including some that are unique to them (think store brands) and designed based on the data that they gather about their customer base's style. However, this has limited the breadth of the assortment they can offer. With active client growth expected to rise, the company wants to expand their selection to satisfy new customers. An expanded assortment will also be important to support strategies implemented in the past year such as Direct Buy, Shop Your Look, and Shop New Colors, as well as Stitch Fix's diversification into men's, kid's and plus size clothing options.

Stitch Fix is mitigating some of its inventory risks by shifting to a "multi-inventory" model in 2021 that includes vendor-managed inventory. With a vendor-managed system, the retailer shares its demand and inventory data with the supplier, which uses that data to replenish the company's inventory when it's running low. It basically puts inventory management on autopilot, leading to a faster supply chain and better availability for in-demand items. A vendor-managed model will allow Stitch Fix to expand or shrink its inventory based on demand and reduce the risk of markdowns from carrying too much merchandise. This is particularly powerful in the hands of a tech-focused business like Stitch Fix, one whose entire business strategy relies on collecting data from customers about their likes and dislikes to deliver a personalized shopping experience. Moving to a multi-inventory model will enable the company over time to meaningfully expand selection and attract more clients – driving higher demand and accelerating growth.

For the brands that partner with Stitch Fix, there are two significant benefits that create a win-win relationship between Stitch Fix and its vendors, ensuring Stitch Fix's ability to successfully implement a vendor-managed inventory model (relative to its competitors) to support its growth while eliminating the need to ramp up its investment in inventory.

First, with the data generated from Stitch Fix, brands are able to improve and evolve their products, based on invaluable and specific feedback about fit, sizing, and design. Each brand has its own 'vendor portal' which enables them to access feedback on how products sold through the service are being received. According to Stitch Fix, 80% of customers leave feedback on returned items, which is testament to the level of investment that customers have in the service. For brands, this experience is different to the process of selling in physical stores, where they are typically left in the dark about why a product doesn't sell. For example, fashion brand Threads 4 Thought made changes to the size of a

button on a particular shirt (based on feedback), eventually resulting in an increase in sales through Stitch Fix as well as other channels. Unsurprisingly, many retail brands are getting on board with Stitch Fix. There are now more than 1,000 partnering brands in the U.S.

Second, vendors can increase demand for their own products through Stitch Fix. Consider the element of discovery that comes along with purchasing clothing through

Stitch Fix. As customers are generally unaware of what they're going to receive, there's a certain amount of surprise and delight involved, with the potential for people to discover new brands or styles that they've never heard of before. This discovery element is also a big benefit for the brands that partner with Stitch Fix, with customers' awareness potentially translating into direct sales or loyalty to individual brands in the long run. Indeed, brand partnerships (whether in terms of sales channels or marketing campaigns) can allow companies to benefit from a combined power, which in turn helps them build awareness and reach a wider audience.

Chapter 7. Preparing a Persuasive Presentation

The final milestone of the Capstone Project is the delivery of your final presentation. Your team will formally present its conclusions and recommendations to a panel of professionals on the final morning of the Bridge program. In corporate valuation, the 'true' value of a company is inherently unknown. So, remember, the goal is to have a defensible point of view and to be persuasive!

What can I expect during the final presentation?

You will have up to 20 minutes to present your valuation, leaving 10 minutes for Q&A immediately thereafter. Your team will play the role of an outside advisor (independent valuation firm, not advocates). Business formal attire is required.

The professionals in your audience are your client and will play the role of a financial buyer. They have hired each Bridge team to value a company for them. They will include professionals from major investment banks, international consulting firms, entrepreneurs, and a member of the Tuck faculty. As potential buyers of the target public company that your team analyzed, your evaluators will want to know: 1) your value estimate of the company's equity, 2) key value drivers that are critical to your estimate, and 3) suggestions as to how a new owner might improve the company's performance should they acquire (and have control over) the company.

What is the process for preparing for the final presentation?

The process of preparing for your final presentation will formally begin during the Business Communications course during the Bridge program. The Business Communication Professor will provide guidance on how to structure your work around a key question (what is the value of the company), structuring your findings and insights around key value drivers, and tightening your recommendations so insights directly affect the valuation. You will learn how to design and structure a PowerPoint slide deck to deliver your message effectively and concisely including use of an Executive Summary, clear headlines, clean horizontal and vertical logic. To support the verbal delivery of your slides, you will learn a framework called HOP-T: headline, orient, key points, and transitions. Your final presentation will put into action the skills you learn in the classroom with Business Communication Professor.

In addition to structured class time in Business Communications, you will have two opportunities to practice your presentation skills prior to the final presentation. First, each team will meet one-on-one with Business Communication Professor to review and refine your Executive Summary. The Executive Summary is the first, and more

important slide in your presentation. It provides your client audience with BLUF (bottom line up front) and is used repeatedly in the presentation to return to and reinforce your valuation and leading value drivers. Second, each team will have the opportunity to deliver their presentation in a rehearsal format the day before the final presentation. During presentation rehearsals you will have up to 20 minutes to deliver your presentation to your peers and a member of the Tuck faculty. You'll receive feedback on your presentation content and delivery. Our collective goal is to take your presentation to the next level,

providing specific areas for improvement. This will give you time to polish your slides and delivery before the big day!

Note that your Bridge Fellow is also a valuable resource in preparing your final presentation. They will guide you all the way from the beginning of your company analyses to the final presentation, helping you stay focused on structure and clarity. Past examples of presentation decks and videos are available in Canvas.

What are some key recommendations for structuring a persuasive presentation?

There are three key recommendations for the final presentation. Note that your Bridge Fellow and Business Communications Professor will offer several additional tips for your final day as and when you move through your preparation.

- First, know your company – e.g., its industry, competitors, strategy, historical performance, etc. Understand your company's objectives and any major forces affecting its ability to achieve those objectives.
- Second, understand the value narrative for your company and tie them directly to the assumptions and value drivers that underlie your value estimate of its equity (e.g., revenue growth, profit margins, investment). See Chapter 6 for some examples. This will require a good balance of company background, financial and market data, as well as your interpretation of what your research uncovers.
- Finally, make a recommendation. It is not enough to perform an extensive company analysis. In the end your team must decide what you think the company is worth. It will make sense to include sensitivity analyses for any assumptions to which your value estimate is sensitive, but you should be firm about your estimate and have a story that backs it up.

What are the most significant challenges to overcome when preparing the final presentation?

Challenges surrounding the final presentation will present themselves related to content, structure, and delivery including, but not limited to, the following:

- In terms of content, be careful not to deliver what looks like a book report. Remember that you are working for a client that is looking for a recommendation regarding the value of a company that they are considering as an acquisition target. This means that everything you cover should be tied directly to the fundamental drivers of value for your company. While your work before the presentation builds to a single number that is the company valuation, the 'story' to your client begins with this number and supports it with your insights and analyses. Focus on tightening your 'story' to meet the needs of your client, including recommendations on ways to improve value and sensitivity analyses revealing the most sensitive aspects of your value estimate.
- Regarding structure, strive to create presentation slides with both horizontal and vertical logic where the slide title is descriptive and the content on each slide reinforces the title. Avoid data without insights and avoid telling a narrative that lacks consistency or logical progression.
- Regarding delivery, it is challenging to divide and conquer among team members. Focus on who is most comfortable telling different parts of the story and focus on transitions so that the presentation appears seamless to the client. In the best presentations, the slides and delivery

tell a complete story, with each person's contribution adding something to that story. A consistent look and feel reinforce that this is a team, not an individual, effort.